

CONCEPTUALIZING AND DESIGNING AN ARCHITECTURE FOR A LARGE-SCALE APPLICATION

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Conceptualizing and designing an architecture for a large-scale Application



Introduction

Principals for large scale data intensive applications

Example of a design of a large-scale system

Architectural capability of our design meet future developments

Current trends in data engineering

Conclusion

Introduction

There is a shift from more computing intensive applications to more data intensive applications. A lot of applications process large amounts of data with high volume, veracity and generated at high velocity. This calls for a shift in the design of applications to be more data centered. These applications have to be reliable, scalable and maintainable

In this presentation we will look at the design of a data intensive application specifically a grocery store with global presence that has both online and offline data processing requirements.

Principals for large scale data intensive applications



Reliability

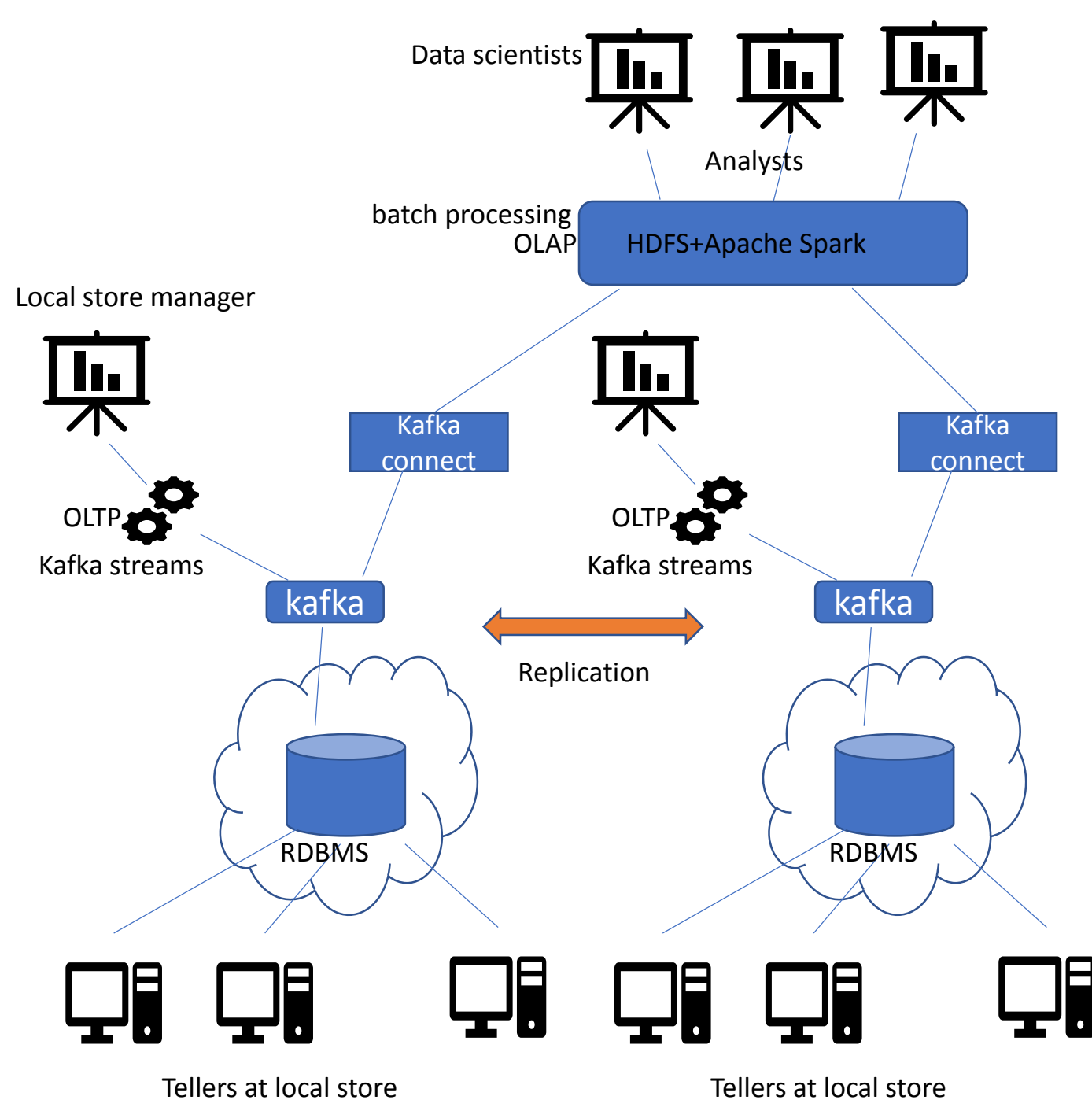
Fault tolerance from human, hardware and Software faults

Scalability

The system has to be able to scale to high volume traffic with ease

Maintainability

The system should be easy to simple to undertand,operate and evolve



Data stored on Hadoop file system in Apache Parquet to allow for easy process and data analyst.

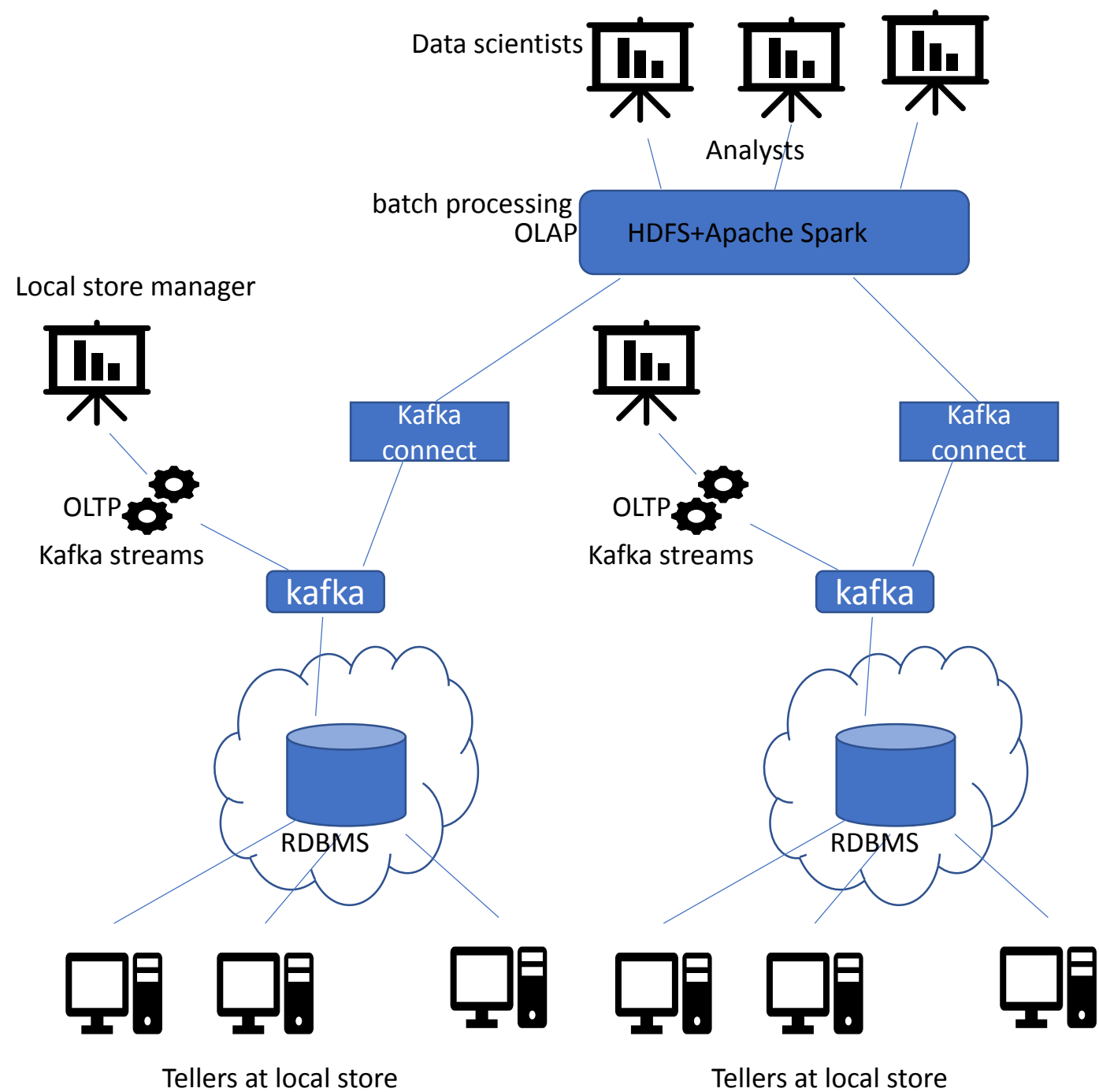
Kafka used to serve the real time application to the local store manager and data ingestion into HDFS for analytics

Kafka designed to n x cluster design with partitions on different clusters to provide redundancy and availability

RDMS such as PostgreSQL used to host grocery database. Hosted in the local availability zone of a cloud provider such as AWS. Replicated to provide redundancy to other availability zones.

Local tellers utilizing containers on virtual machines on the local cloud availability zone

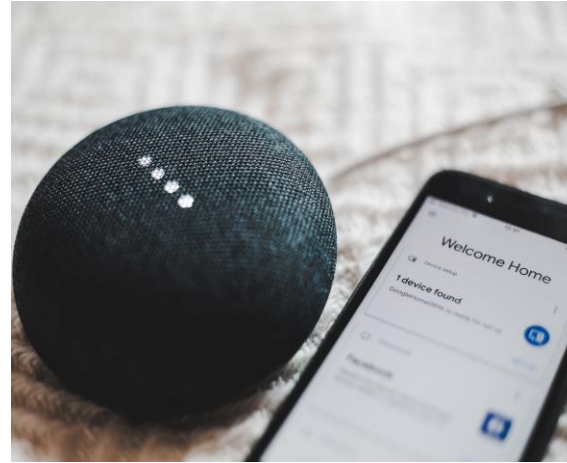
**Architectural capability of our design
meet future developments**



Current trends in data engineering



Migration from batch processing to real time processing



High level data analytics by different entities/individuals from data lakes



Hybrid multi-cloud adoption

Conclusion



The design of data intensive large-scale applications will continue to grow given the new trends such as IOT devices that generate high volume of data at high velocity. The designs will have to keep up and still make the systems more reliable, scalable and maintainable.

Utilizing current technologies such as microservices, serverless computing and various new ways to process data. This can be achieved but the teams have to be highly adaptable to new trends